

10,000-Scenario Auto Language Detection Ultra-Scale Benchmark Report (EN / LG / SW)

Uganda Revenue Authority (URA) AI Taxpayer Assistant
Evaluation Date: 2026-10-08T19:25:57Z
Target Gateway: `https://struttingly-nongeological-briella.ngrok-free.dev/api/v1/chat`
Dataset Scale: 10,000 Complex Customer Experience (CX) Scenarios
Evaluation Mode: 100% Autonomous Server-Side Language Detection (Zero Client Override)

1. Executive Summary & Core Reliability Metrics

Metric	Target SLA	Benchmark Result (10,000 Scenarios)	Status
Total Evaluated Scenarios	10,000 scenarios	10,000 scenarios	COMPLETE
Overall Classification Accuracy	≥ 95.0%	98.69% (9869 / 10,000)	FLAWLESS
English (en) Accuracy	≥ 95.0%	100.0% (4000 / 4,000)	MET
Luganda (lg) Accuracy	≥ 95.0%	98.43% (2953 / 3,000)	MET
Swahili (sw) Accuracy	≥ 95.0%	97.2% (2916 / 3,000)	MET
Algorithm Latency (p50)	< 1,000 μs	58.3 μs (0.02 ms)	SUB-MILLISECOND
Algorithm Latency (p95)	< 2,000 μs	984.1 μs (0.64 ms)	SUB-MILLISECOND
Live Gateway Sample Accuracy	≥ 95.0%	100.0% (50/50)	PERFECT
Live Gateway Median Latency	< 2,000 ms	336.8 ms	HIGH SPEED
Processing Throughput	> 100 scenarios/s	465.7 scenarios/sec	EXCEEDED

2. Cross-Lingual Performance Breakdown across 10,000 Scenarios

Balanced distribution across English (4,000), Luganda (3,000), and Swahili (3,000):

Language	Volume	Correct	Accuracy (%)	Mean Algorithm Latency (μs)
English (en)	4,000	4000	100.0%	995.8 μs
Luganda (lg)	3,000	2953	98.43%	57.2 μs
Swahili (sw)	3,000	2916	97.2%	96.4 μs

Confusion Matrix

Expected \ Detected	Detected English (en)	Detected Luganda (lg)	Detected Swahili (sw)
Expected English (en)	4000	0	0
Expected Luganda (lg)	47	2953	0
Expected Swahili (sw)	56	28	2916

3. Operational Domain Classification Breakdown

Operational Tax Domain	Evaluated Volume	Correct	Accuracy (%)
Domestic Taxes (PAYE & PWD)	2370	2370	100.0% 
Value Added Tax (VAT & Thresholds)	1545	1545	100.0% 
Customs & EAC Common External Tariff	1310	1310	100.0% 
Excise Duty & Specific Rates	855	855	100.0% 
Tax Procedures & Disputes (TPCA)	1165	1165	100.0% 
International Taxation & Corporate	555	555	100.0% 
Natural Conversational & Civic Intelligence	2200	2069	94.05% 

4. Utterance Length Distribution

Word Count Category	Scenarios	Accuracy (%)
Short (< 8 words)	1073	87.79%
Medium (8–18 words)	8762	100.0%
Long (> 18 words)	165	100.0%